



# Benchmarking and Performance Analysis of LLM Inferencing Systems

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Systems for LLMs
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Workload Picked

Batch of Requests through Various Engines

LLM Request Timelines

Throughput vs Load for vLLM

NSYS Measurements

GPU Memory Breakdown

Experiments with Multiple Families

Broader Goal

# Contents




1. Workload Picked
2. Batch of Requests through Various Engines
3. LLM Request Timelines
4. Throughput vs Load for vLLM
5. NSYS Measurements
6. GPU Memory Breakdown
7. Experiments with Multiple Families
8. Broader Goal

# Workload Setup





- **Goal:** Evaluate inference performance of LLMs under short queries with large shared context.
- **Model:** Qwen 0.5B, Tiny Llama 1.1B, served using vLLM/SGLang/Transformers.
- **Hardware:** NVIDIA RTX6000 48GB GPU.
- **Metrics:** Throughput (req/s), latency, GPU utilization.

| Parameter | Value/Range |
|---|---|
| System Prompt Length | ≈1500 tokens (shared across requests) |
| Query Rate | 5–700 req/sec |
| Query Length | 10–20 tokens |
| Output Length | 30–50 tokens |
| Batch Size | Adaptive |

Table: Workload Characteristics

# Experiment Setup: Engine Comparison





Workload Picked

Batch of Requests through Various Engines

LLM Request Timelines

Throughput vs Load for vLLM

NSYS Measurements

GPU Memory Breakdown

Experiments with Multiple Families

Broader Goal

- Compare inference engines: **Transformers**, **vLLM**, **SGLang**, **NEO**.
- Batch of 24 requests, each sharing the same system prompt.
- Each request: $\sim$20 input tokens $\rightarrow$ $\sim$40 generated tokens.
- Measure **total time taken** to process the batch.

**Hypothesis**

Engines leveraging prefix caching (e.g., SGLang) should perform far better than naive batching in Transformers.

# Batch of requests through various Engines

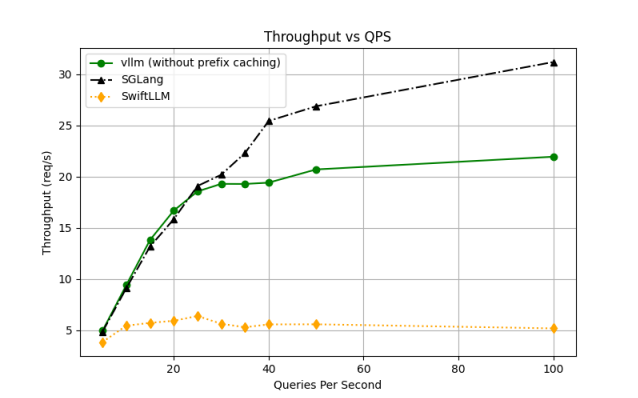



Figure: Throughput comparison across varying workloads

- **SwiftLLM:** lightweight vLLM clone (∼2000 LOC), but doesn't performs that well.
- **SGLang:** achieves best performance via radix-tree prefix sharing.

| Technique Used | Time (s) |
|---|---|
| Transformers (batch = 1) | 30.21 |
| Transformers (batch = 2) | 19.51 |
| Transformers (batch = 3) | 18.64 |
| vLLM (no system prompt pre-filling) | 0.45 |
| SGLang (fork) | 1.03 |
| SGLang (run_batch) | 0.40 |
| NEO | 0.55 |

Table: Measuring the time taken to process a batch of 24 requests using different techniques


Systems for LLMs
Saksham Rathi
Workload Picked
Batch of Requests through Various Engines
LLM Request Timelines
Throughput vs Load for vLLM
NSYS Measurements
GPU Memory Breakdown
Experiments with Multiple Families
Broader Goal

# Experiment Setup: Request Timelines





- Aim: Measure time spent in **prefill** vs **decode** stages for 20 req/sec workload (sending a new request after every 0.05 seconds).
- Using vLLM V0 engine without chunked prefill.
- Tool: Custom logging of per-request timestamps.

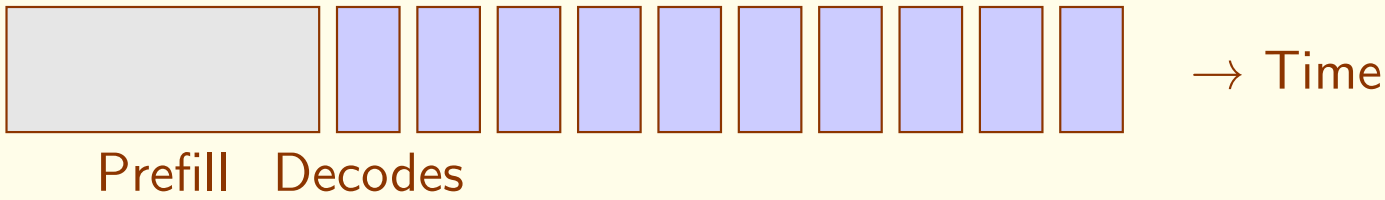



Figure: Simplified per-request timeline

Workload Picked
Batch of Requests through Various Engines
LLM Request Timelines
Throughput vs Load for vLLM
NSYS Measurements
GPU Memory Breakdown
Experiments with Multiple Families
Broader Goal

# Results: LLM Request Timelines

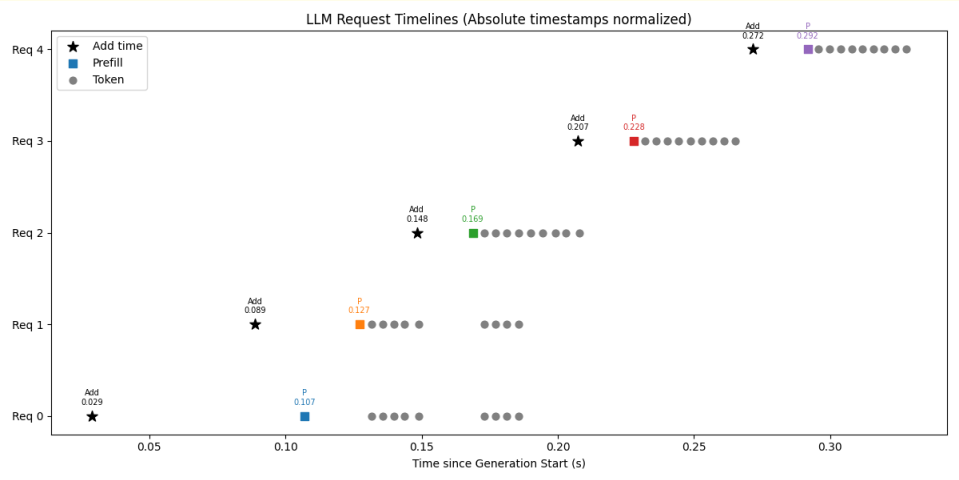



Figure: 20 Requests/sec – No chunked prefill – vLLM V0

- Prefill dominates runtime due to large prompt.
- Prefill and decode cannot overlap here.
- Low TTFT but high total latency.


Systems for LLMs
Saksham Rathi
Workload Picked
Batch of Requests through Various Engines
LLM Request Timelines
Throughput vs Load for vLLM
NSYS Measurements
GPU Memory Breakdown
Experiments with Multiple Families
Broader Goal

# Results: Throughput vs Load for vLLM

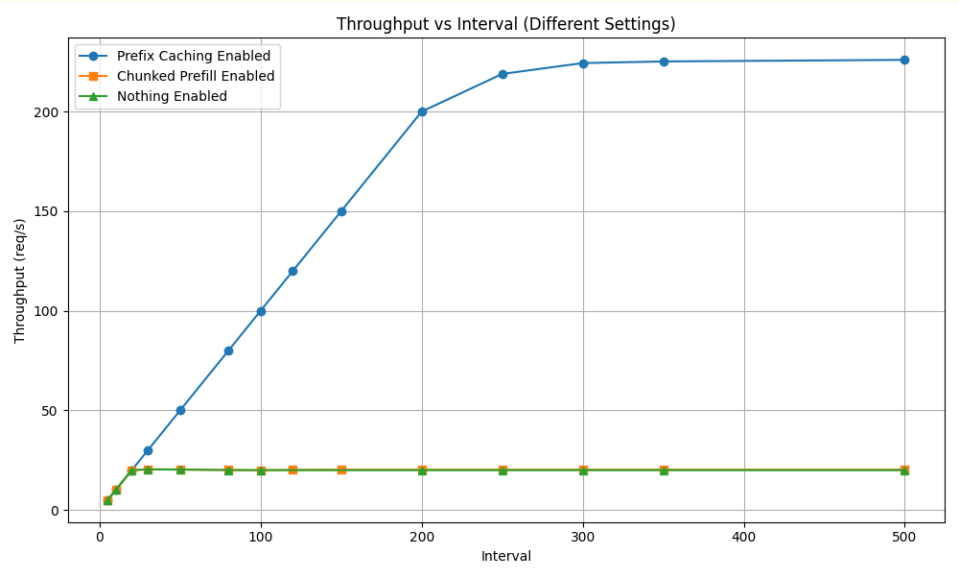



Figure: Throughput vs Load

- Varying the input load, and comparing different modes.
- Chunked prefill: limited benefit for small queries.
- Prefix caching: major boost to throughput.
- Batch tuning further improves efficiency.

**Observation**

Saturation begins at ∼300 req/sec — compute bottleneck.


Systems for LLMs
Saksham Rathi
Workload Picked
Batch of Requests through Various Engines
LLM Request Timelines
Throughput vs Load for vLLM
NSYS Measurements
GPU Memory Breakdown
Experiments with Multiple Families
Broader Goal

# Results: NSYS Measurements

## Goal

Understand compute vs memory utilization and bottlenecks.




Figure: NSYS Metrics Snapshot


Systems for LLMs
Saksham Rathi
Workload Picked
Batch of Requests through Various Engines
LLM Request Timelines
Throughput vs Load for vLLM
NSYS Measurements
GPU Memory Breakdown
Experiments with Multiple Families
Broader Goal

# Observation: NSYS Profiling





- Compute: SM Active and Warp occupancy.
- Memory: vLLM books full GPU memory upfront.
- vLLM scheduler stats show the actual KV Cache Usage.
- PCIe Bandwidth low $\rightarrow$ small workloads, minimal CPU transfer.

# GPU Memory Breakdown

| Component | Memory Occupied |
|---|---|
| Weights | 0.93 GiB |
| Peak Activation | 4.07 GiB |
| Non-Torch | 0.02 GiB |
| CUDA Graph | 0.41 GiB |
| KV Cache: Intersection (Prefix) | 1,280 tokens / 0.0157 GiB |
| KV Cache: Extras for Requests | 640 tokens (10 requests × 64 tokens) / 0.0078 GiB |
| KV Cache: Free | 2,672,880 tokens / 32.8445 GiB |

Table: GPU Memory Breakdown and KV Cache Usage

# Observation: Memory Profiling





Workload Picked

Batch of Requests through Various Engines

LLM Request Timelines

Throughput vs Load for vLLM

NSYS Measurements

GPU Memory Breakdown

Experiments with Multiple Families

Broader Goal

- KV Cache footprint is tiny — prefix reuse highly memory-efficient.
- Most GPU memory remains free $\rightarrow$ headroom for larger batches.
- Suggests potential for multi-family prefix caching in future work.

# Generation of these families

- Let's say we already have a large system prompt of length 1500 tokens.
- We generate multiple families of requests by prepending a unique family identifier to the system prompt.
- For example, for family 1, we prepend "AA", for family 2, "AB", and so on.
- This ensures that although the families have a large substring in common, but their prefixes always differ, and the KV Cache block hashes won't match.
- This in turn implies, that we need to compute and store separate KV Cache blocks for each family.

# Experiments with Multiple Families

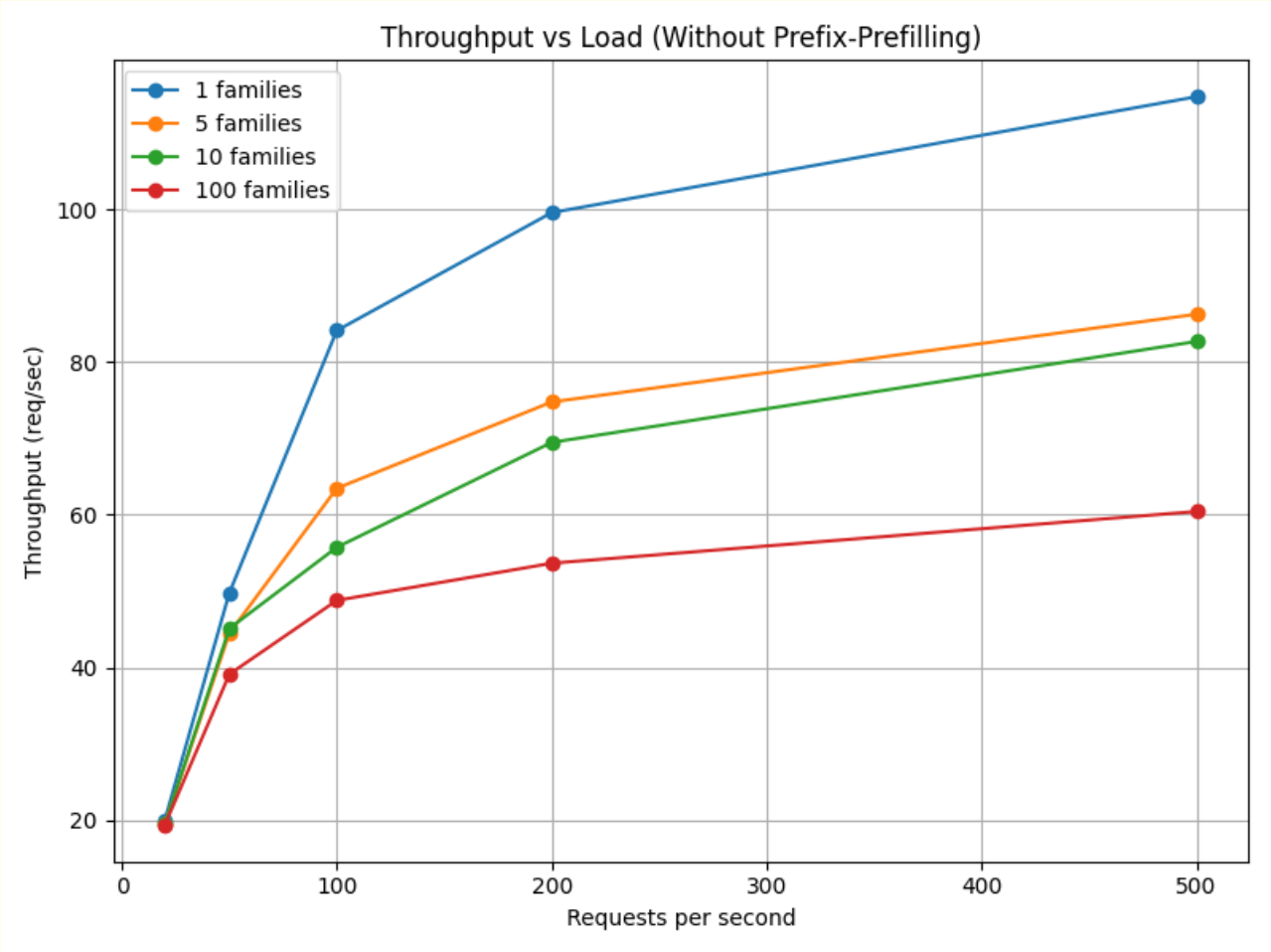



Figure: Throughput vs Load without prefix pre-filling


Systems for LLMs

Saksham Rathi

Workload Picked

Batch of Requests through Various Engines

LLM Request Timelines

Throughput vs Load for vLLM

NSYS Measurements

GPU Memory Breakdown

Experiments with Multiple Families

Broader Goal

# Experiments with Multiple Families

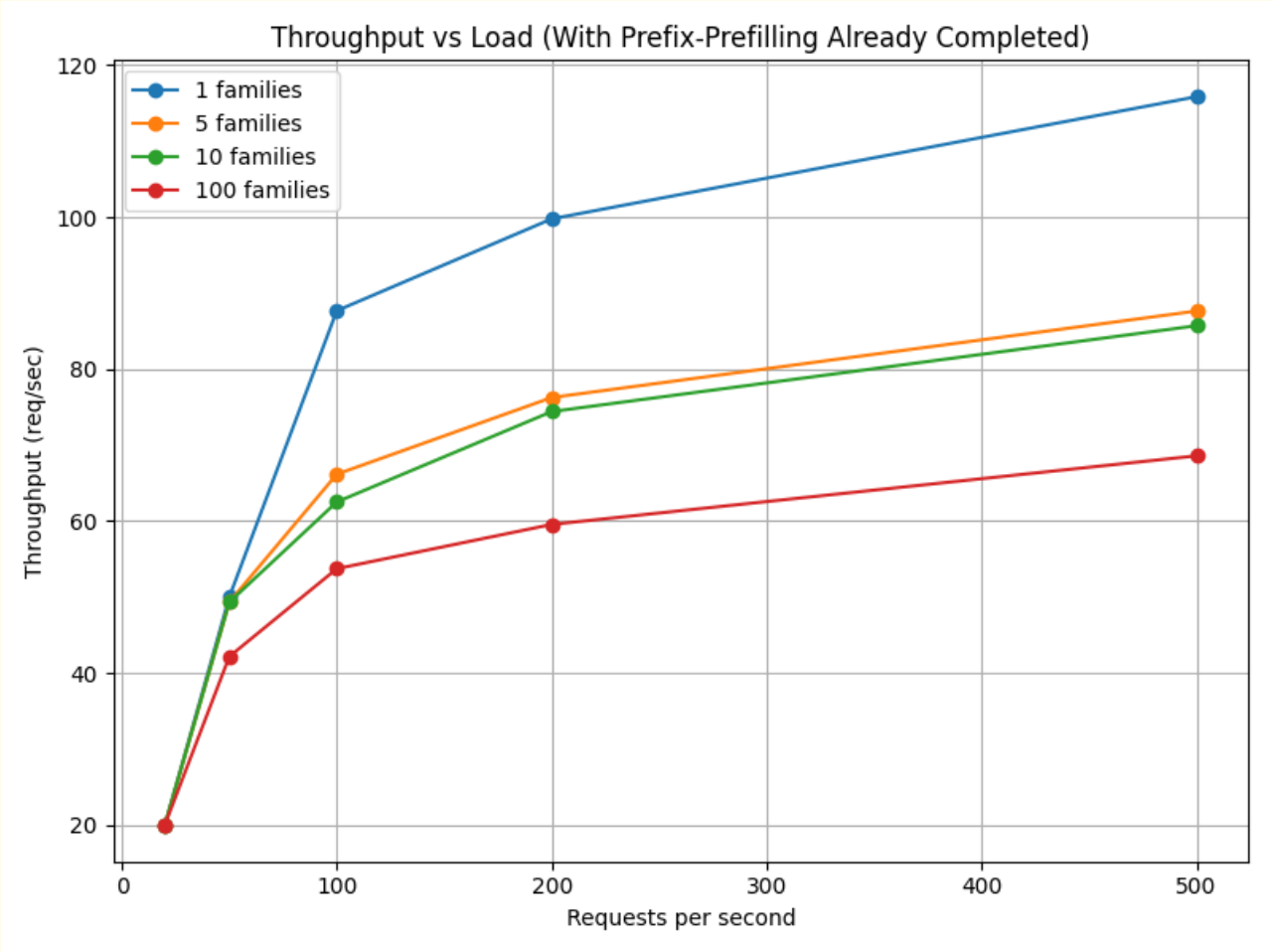



Figure: Throughput vs Load with prefix pre-filling


Systems for LLMs
Saksham Rathi
Workload Picked
Batch of Requests through Various Engines
LLM Request Timelines
Throughput vs Load for vLLM
NSYS Measurements
GPU Memory Breakdown
Experiments with Multiple Families
Broader Goal

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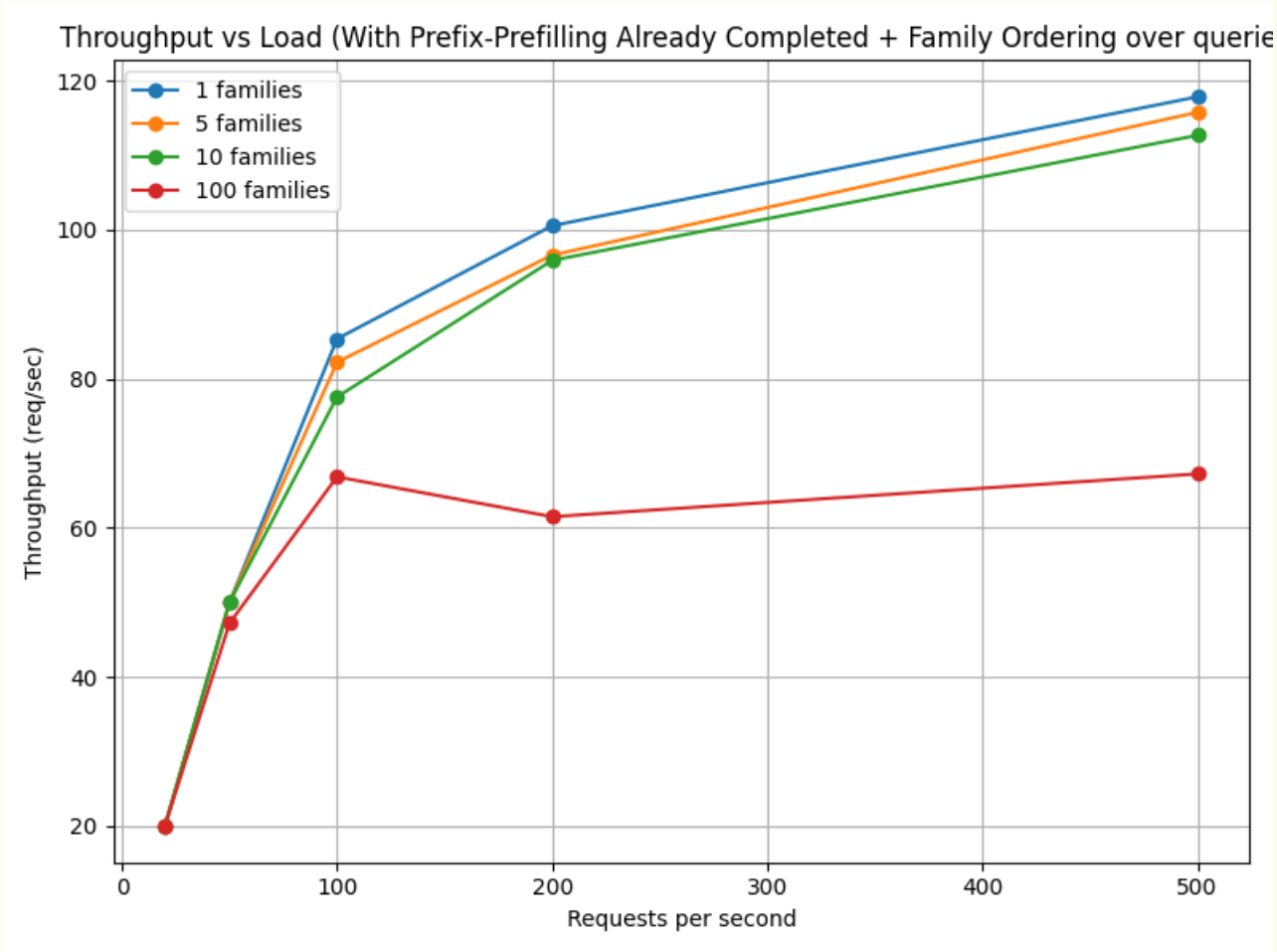



Figure: Throughput vs Load with prefix pre-filling with ordered family IDs


Systems for LLMs
Saksham Rathi
Workload Picked
Batch of Requests through Various Engines
LLM Request Timelines
Throughput vs Load for vLLM
NSYS Measurements
GPU Memory Breakdown
Experiments with Multiple Families
Broader Goal

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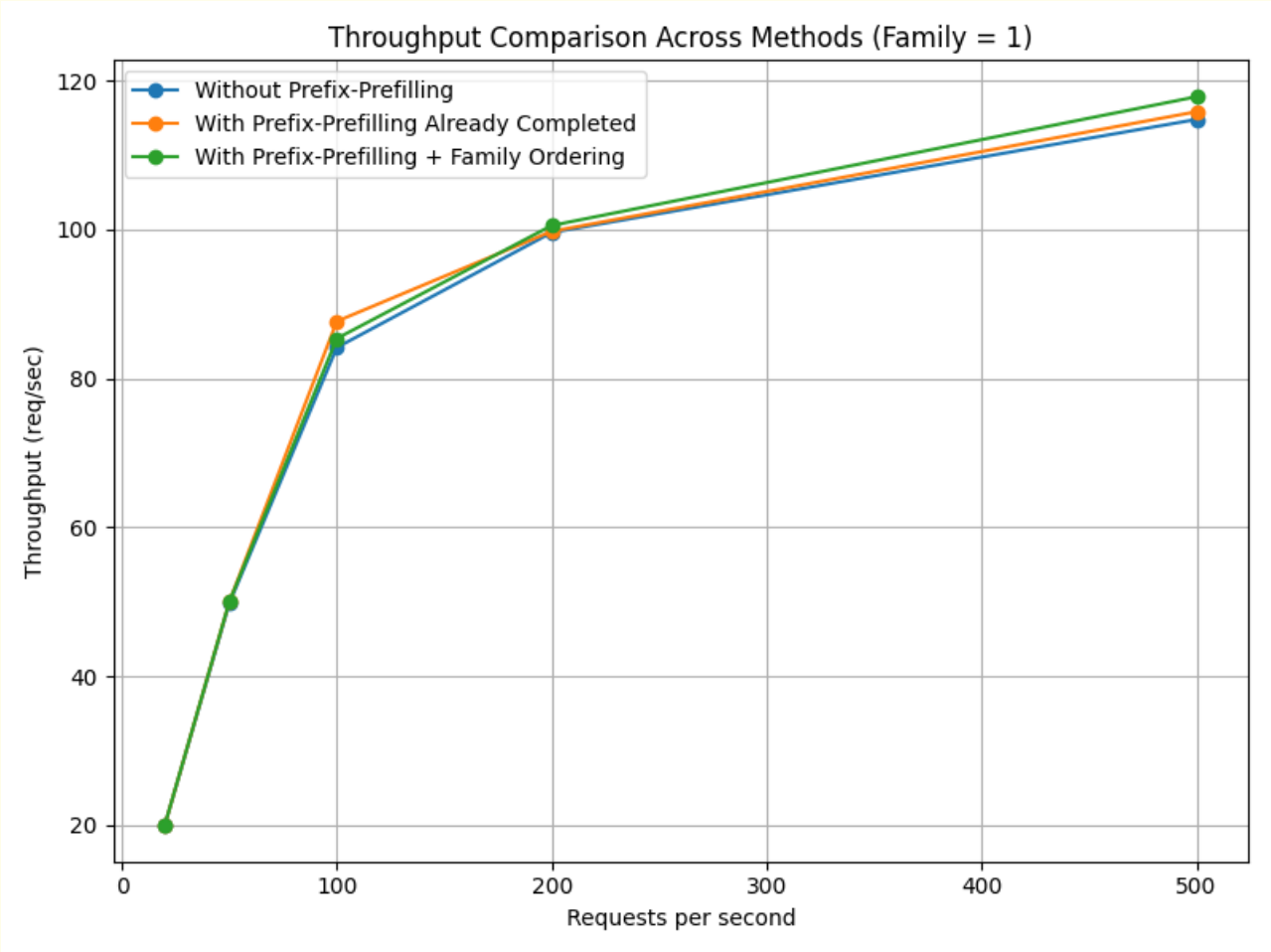



Figure: Throughput vs Load for 1 family


Systems for LLMs
Saksham Rathi
Workload Picked
Batch of Requests through Various Engines
LLM Request Timelines
Throughput vs Load for vLLM
NSYS Measurements
GPU Memory Breakdown
Experiments with Multiple Families
Broader Goal

# Experiments with Multiple Families

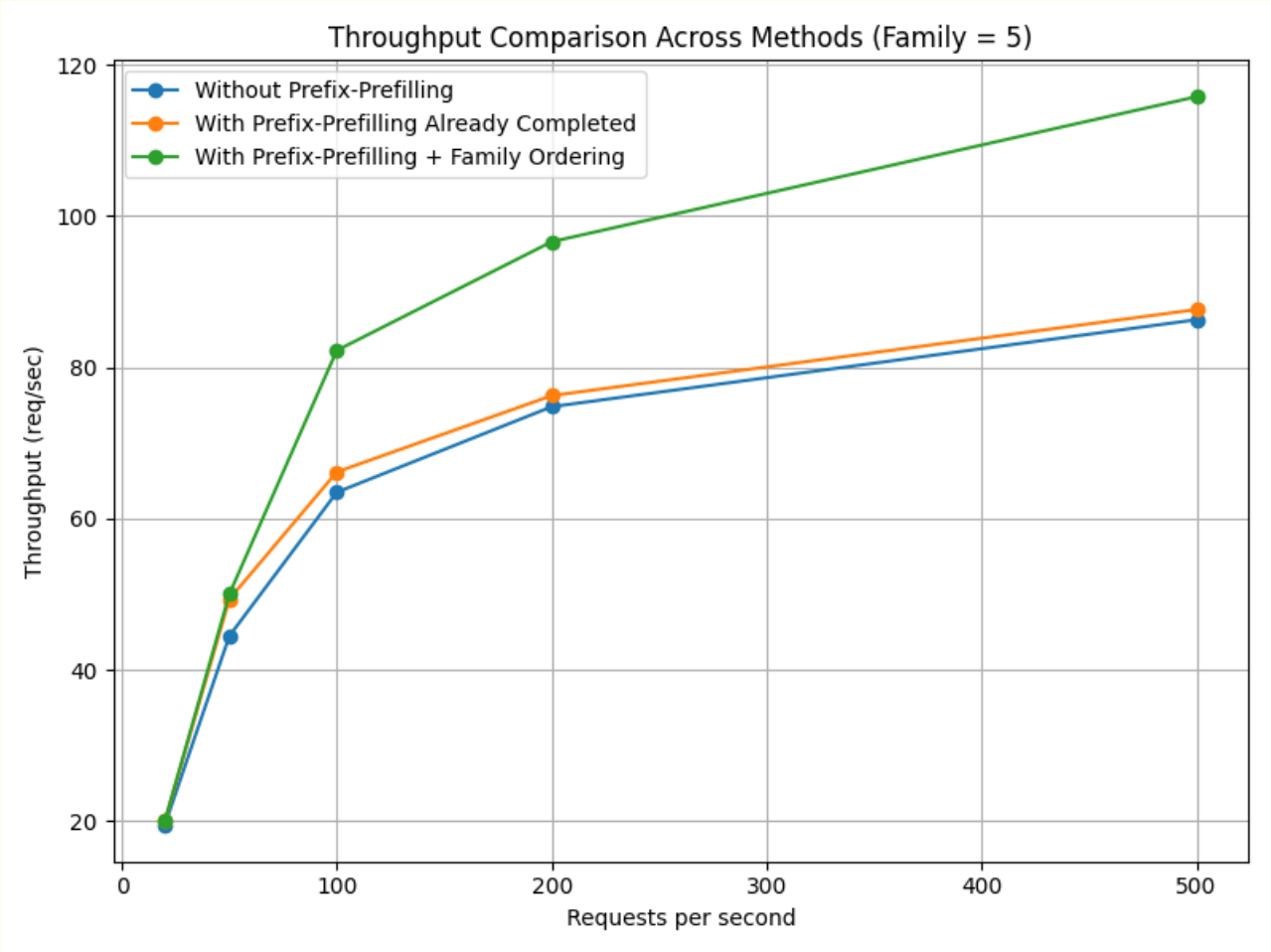



Figure: Throughput vs Load for 5 families


Systems for LLMs
Saksham Rathi
Workload Picked
Batch of Requests through Various Engines
LLM Request Timelines
Throughput vs Load for vLLM
NSYS Measurements
GPU Memory Breakdown
Experiments with Multiple Families
Broader Goal

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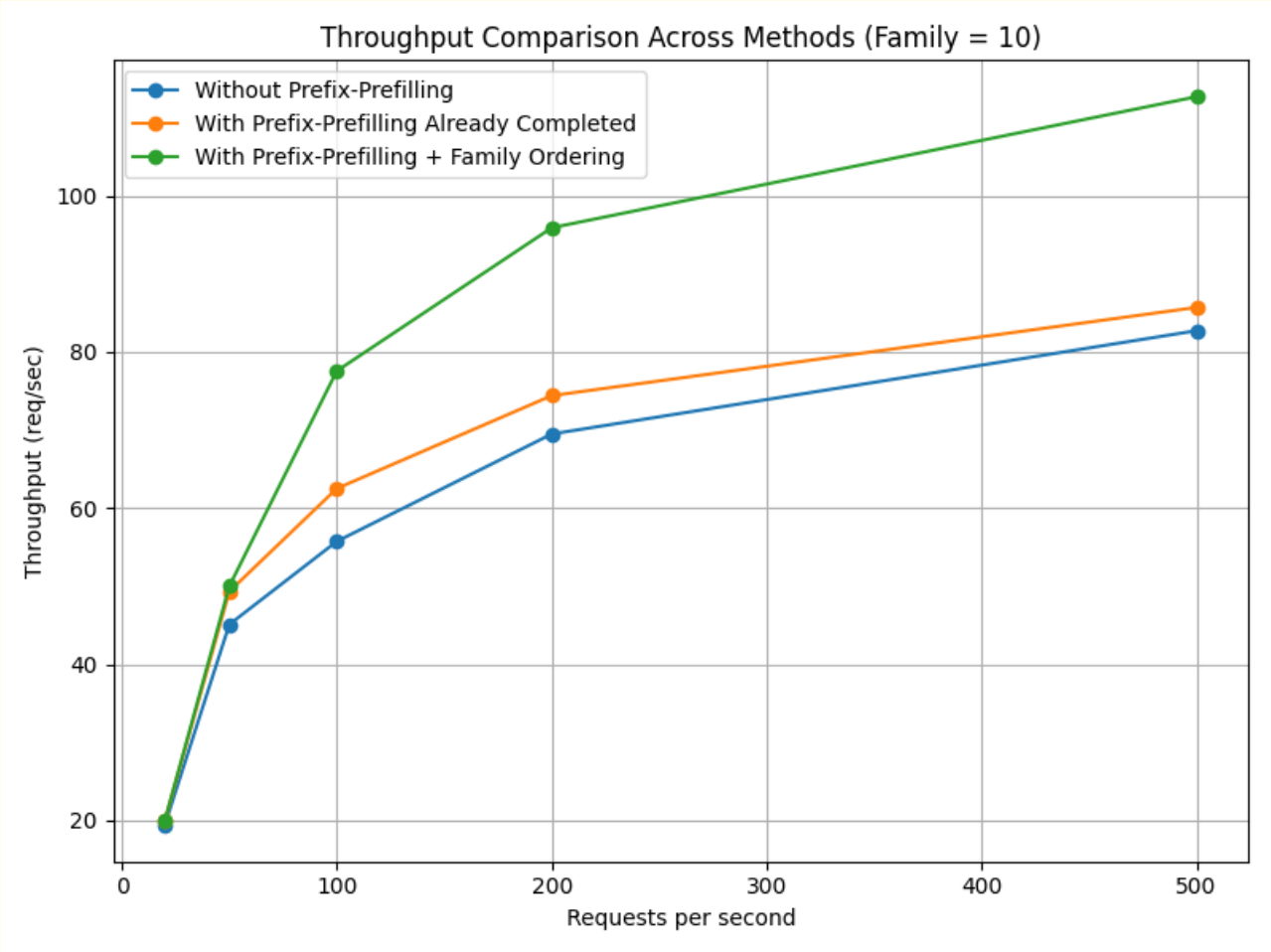



Figure: Throughput vs Load for 10 families


Systems for LLMs
Saksham Rathi
Workload Picked
Batch of Requests through Various Engines
LLM Request Timelines
Throughput vs Load for vLLM
NSYS Measurements
GPU Memory Breakdown
Experiments with Multiple Families
Broader Goal

# Experiments with Multiple Families

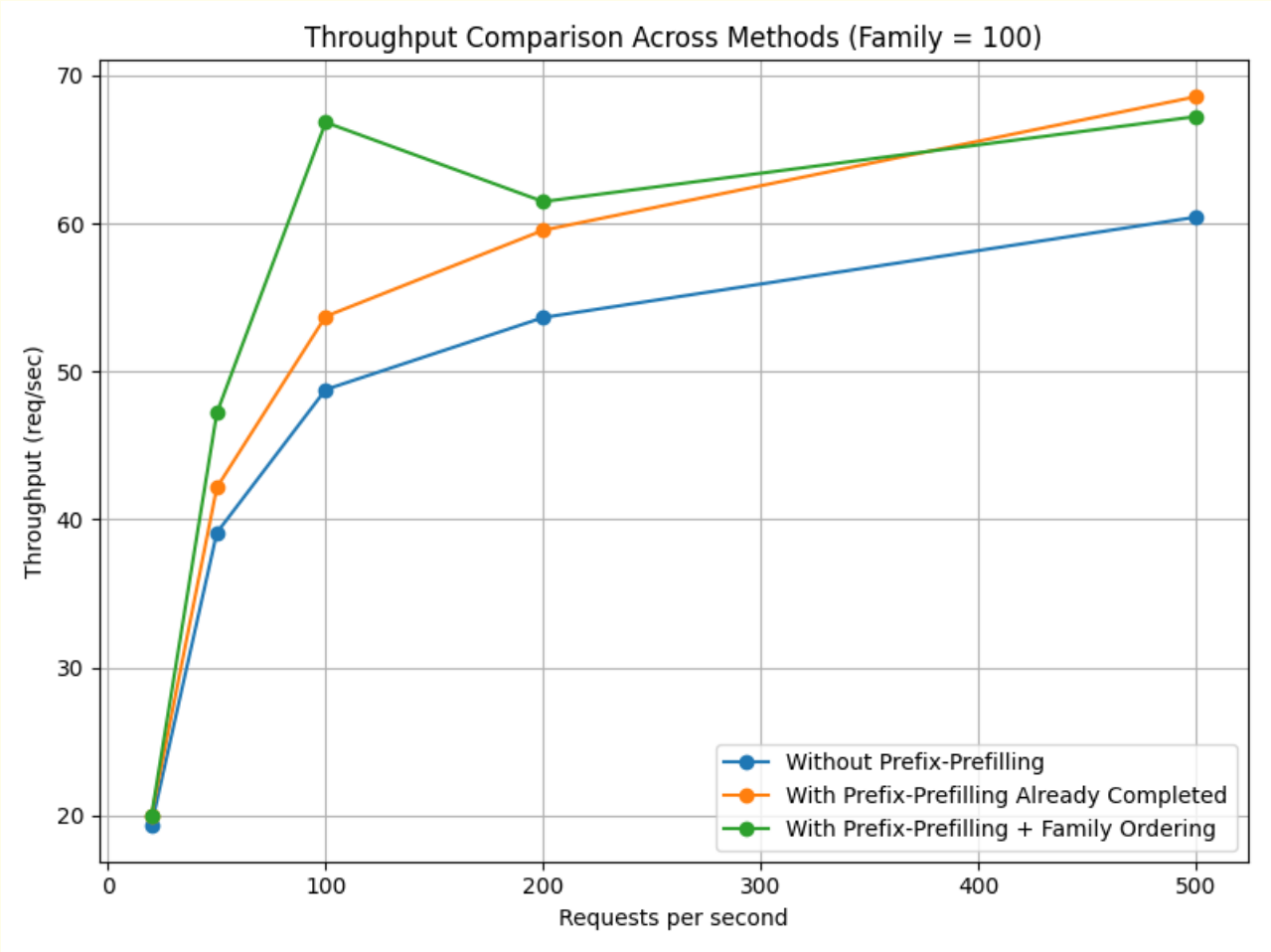



Figure: Throughput vs Load for 100 families


Systems for LLMs
Saksham Rathi
Workload Picked
Batch of Requests through Various Engines
LLM Request Timelines
Throughput vs Load for vLLM
NSYS Measurements
GPU Memory Breakdown
Experiments with Multiple Families
Broader Goal

# Broader Goal: Motivation





Workload Picked

Batch of Requests through Various Engines

LLM Request Timelines

Throughput vs Load for vLLM

NSYS Measurements

GPU Memory Breakdown

Experiments with Multiple Families

Broader Goal

- **Without prefix sharing:** batch limited by model context length (e.g., 32,768 tokens for a small Qwen model $\rightarrow$ 1500 tokens $\times$ 20 requests).
- **With prefix sharing:** utilization-limited, not context-limited. As the number of concurrent requests grows, per-request latency increases, and beyond a point, overall throughput would start declining.

**Next Step**

Design a scheduler for families of such requests. At each scheduling step, we must decide whether to prefill a shared prefix or decode tokens for an existing request.

- Explore KV cache offloading to CPU for extended workloads.

# Future Directions: Quality vs Performance





- Strategy 1: Pass full system prompt each time.
- Strategy 2: Fine-tune for families of similar requests.
- Study trade-offs:
  - **Performance:** latency, throughput, memory use.
  - **Quality:** output accuracy, coherence.





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Batch of Requests through Various Engines

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Experiments with Multiple Families

Broader Goal

# Thank You